

Protection against cancer by dietary IP6 and inositol.

Inositol hexaphosphate (IP(6)) is a naturally occurring polyphosphorylated carbohydrate, abundantly present in many plant sources and in certain high-fiber diets, such as cereals and legumes. In addition to being found in plants, IP(6) is contained in almost all mammalian cells, although in much smaller amounts, where it is important in regulating vital cellular functions such as signal transduction, cell proliferation, and differentiation. For a long time IP(6) has been recognized as a natural antioxidant. Recently IP(6) has received much attention for its role in cancer prevention and control of experimental tumor growth, progression, and metastasis. In addition, IP(6) possesses other significant benefits for human health, such as the ability to enhance immune system, prevent pathological calcification and kidney stone formation, lower elevated serum cholesterol, and reduce pathological platelet activity. In this review we show the efficacy and discuss some of the molecular mechanisms that govern the action of this dietary agent. Exogenously administered IP(6) is rapidly taken up into cells and dephosphorylated to lower inositol phosphates, which further affect signal transduction pathways resulting in cell cycle arrest. A striking anticancer action of IP(6) was demonstrated in different experimental models. In addition to reducing cell proliferation, IP(6) also induces differentiation of malignant cells. Enhanced immunity and antioxidant properties also contribute to tumor cell destruction. Preliminary studies in humans show that IP(6) and inositol, the precursor molecule of IP(6), appear to enhance the anticancer effect of conventional chemotherapy, control cancer metastases, and improve quality of life. Because it is abundantly present in regular diet, efficiently absorbed from the gastrointestinal tract, and safe, IP(6) + inositol holds great promise in our strategies for cancer prevention and therapy. There is clearly enough evidence to justify the initiation of full-scale clinical trials in humans.